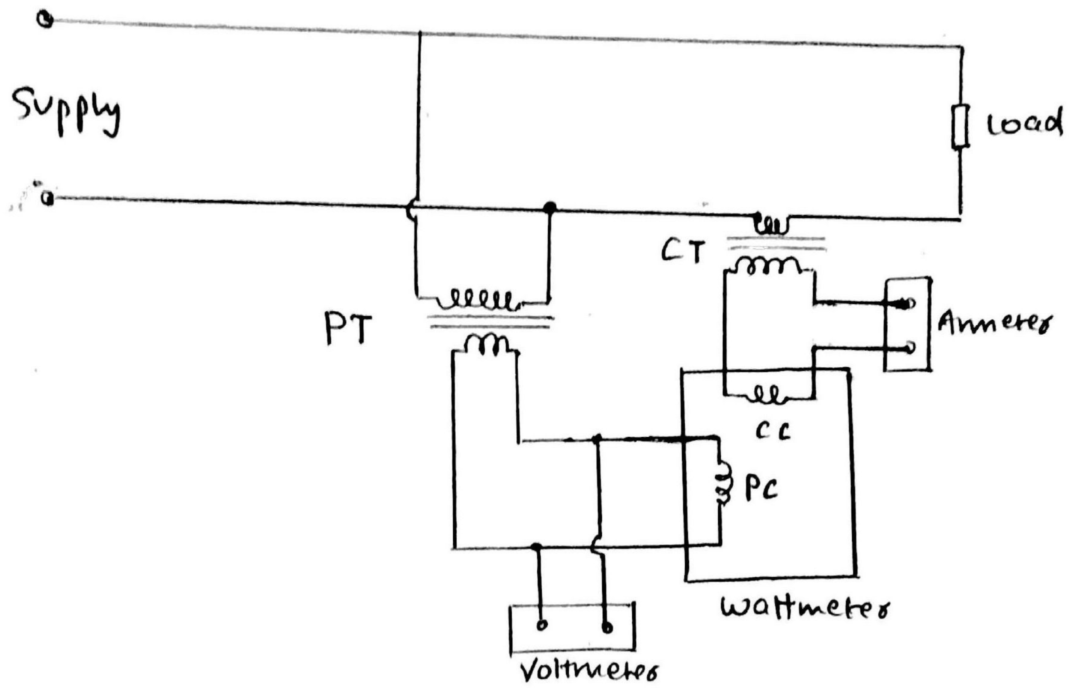
00711504915

MEASUREMENT OF POWER USING INSTRUMENT TRANSFORMERS

Instrument Transformers are used for extension of instrument range. High voltages and high currents can be stepped down to values which are convinient for Measurement. Hence, these Instrument Transformers can be used for 'Measurement of Power'.



→ The Primary winding of C.T. is connected in series with load and the secondary winding of C.T. is conneted in series with an ammeter and the current coil of wattmeter.

→ The Primary winding of P.T. is connected accross the supply and the secondary windings of P.T. are connected to the potential coil and voltmeter in parallel.

This arrangement helps in measuring the power easily as CT and PT help in stepping down value of current and voltage in C.C. and P.C. respectively.

Voltmeters and Ammeters are effected by Ratio errors and wattmeters are effected by phase angle errors. Hence connections are made.

Correction Factor

$$K = \frac{\cos\phi}{\cos\alpha \cos\beta}$$

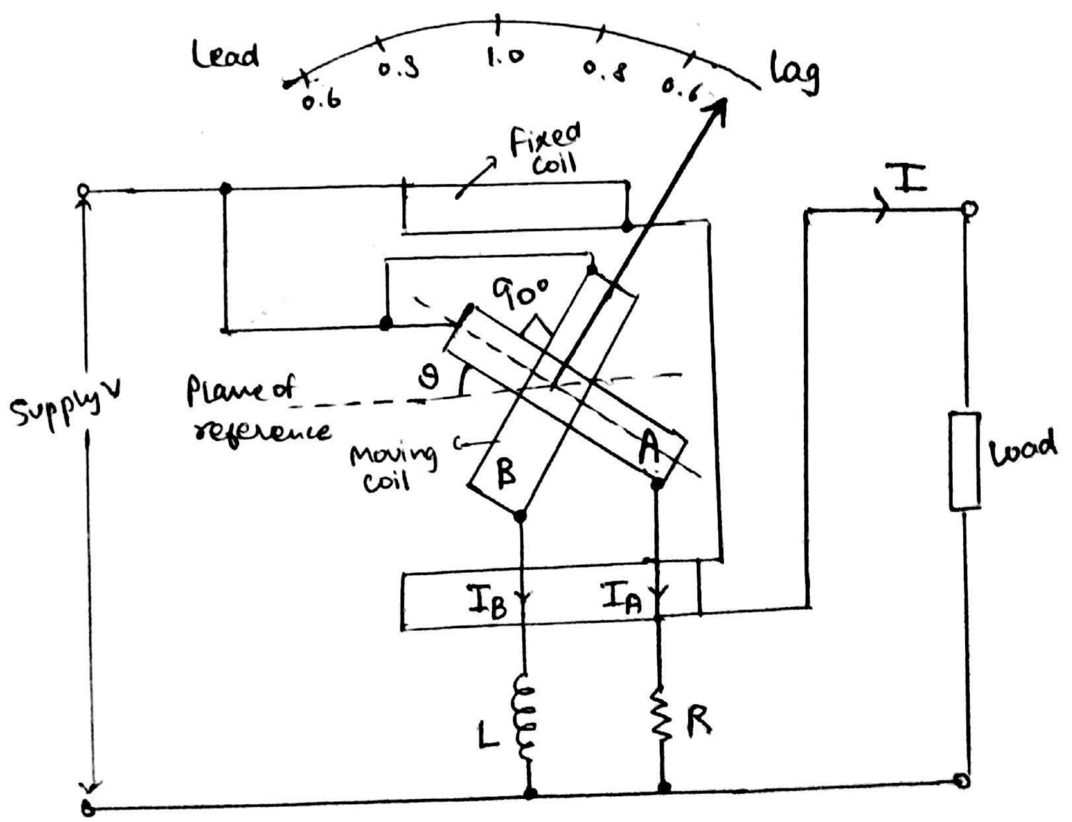
where ϕ is phase angle between current and voltage, α is phase angle between currents in C.C. and P.C. and β is angle by which current in P.C. lags voltage accross secondary winding of P.T.

Power = K × actual Ratio of P.T. × actual ratio of C.T. × wattmeter reading

Power Factor Meters $\left(\cos\phi = \frac{P}{VI}\right)$

Power factor meters are used to indicate the power factors of various circuits accurately. Power factor meters have a current coil and a pressure coil. The current coil (C.C) carries current into the circuit whose power factor has to be measured. The pressure coil or circuit is connected across the circuit whose power factor as to be calculated. The deflection for power factor is indicated by a pointer. Type → Dynamometer type
→ Moving Iron type

I Single Phase Electrodynamometer Type Power Factor Meter.

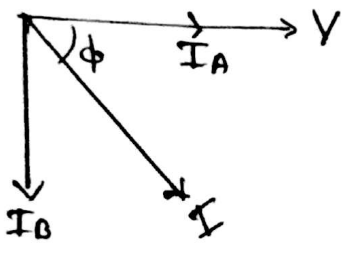


It consists of a fixed coil which acts as a current coil, split into 2 parts, carrying current to the circuit

It consists of 2 identical pressure coils, coil A and B, which are alligned 90° to each other. These pressure coils are the moving coils for the system.

Coil A and B are connected across the supply using a resistor and an inductor respectively.
Coil A has R connected in series with it and Coil B has L connected in series with it.
R, L are chosen such that $\boxed{R = \omega L}$

Now, current I_A and I_B flowing through the coils have different phases.



Torque is produced due to currents in the coils.
Torque produced in coil A and coil B will be in opposite direction due to difference in phase of currents

We know,

$T \propto I' I'' \cos\phi$ and $T \propto \frac{dM}{d\theta}$ where $M = -M_{max}\cos\theta$

(due to interaction of flux of systems) (due to mutual inductance)

$\therefore$ $T_A = k I_A I \cos\phi \, M_{max} \sin\theta$, for coil A

and $T_B = k I_B I \cos(90 - \phi) \, M_{max} \sin(90 + \theta)$, for coil B

[Note: I_A and I_B will eventually be equal in magnitude since they are connected across same voltage and $\underline{R = \omega L}$, hence have same reactance]

At position of equilibrium for the system,

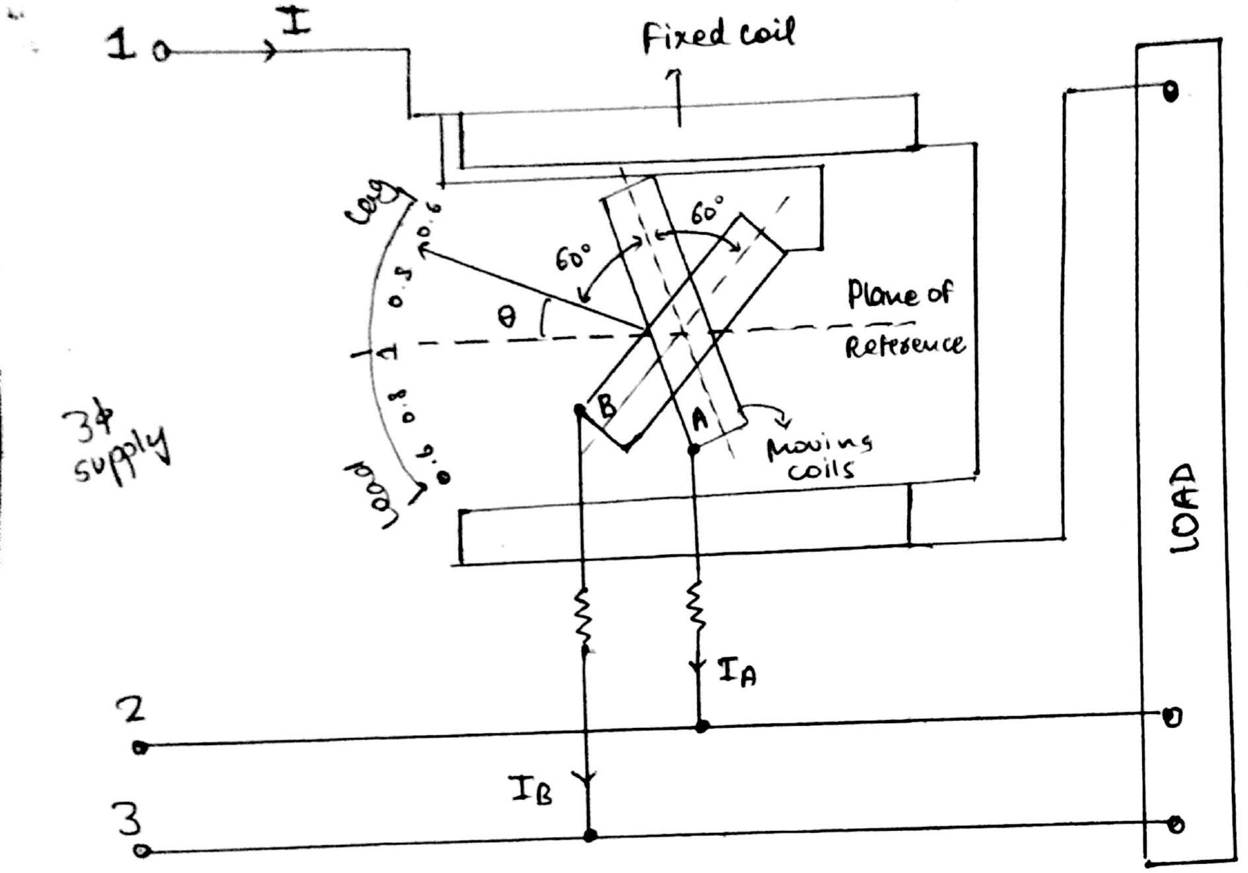
$T_A = T_B$

we get, $\theta = \phi$, is the deflection produced

Hence the angle by which the pointer will be deflected will be the phase angle ϕ of the circuit. Hence $\cos\phi$ is calculated.

($\because$ $R = \omega L$, It is used for constant frequency source or supply.)

II Three Phase Electrodynamometer Power Factor Meter



→ Useful for balanced loads.

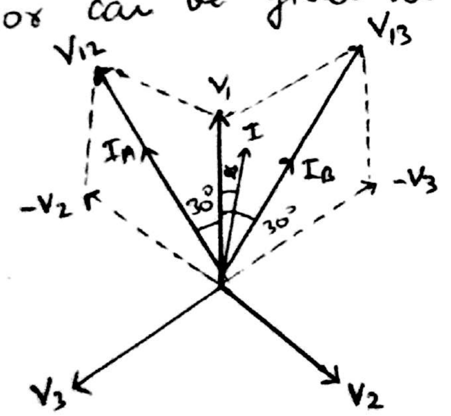
→ phase splitting is not required

→ Indications are frequency independent

In this type of Power factor meter, moving coils are placed across the supply, making an angle of 120° b/w their planes. Both coils A and B are connected in series with a resistor. Coil A is connected b/w phase 1 and 2 and coil B across phase 1 and 3.

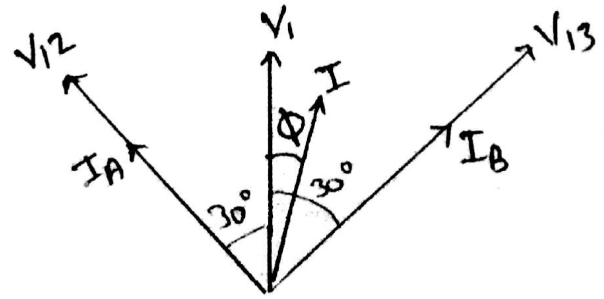
∵ Resistive circuit ∴ I_A and I_B are in phase with voltages V_{12} and V_{13} respectively.

Phasor can be given as-



(I lags V_1 by angle ϕ)

Now, Torque is produced in the moving coils which oppose each other.



In coil A,

$$T_A = K V_{12} I M_{max} \cos(30+\phi) \sin(60+\theta)$$

$$\therefore T_A = \sqrt{3} K V_1 I M_{max} \cos(30+\phi) \sin(60+\theta)$$

In coil B, similarly.

$$T_B = K V_{13} I M_{max} \cos(30-\phi) \sin(120°+\theta)$$

$$\therefore T_B = \sqrt{3} K V_1 I M_{max} \cos(30-\phi) \sin(120°+\theta)$$

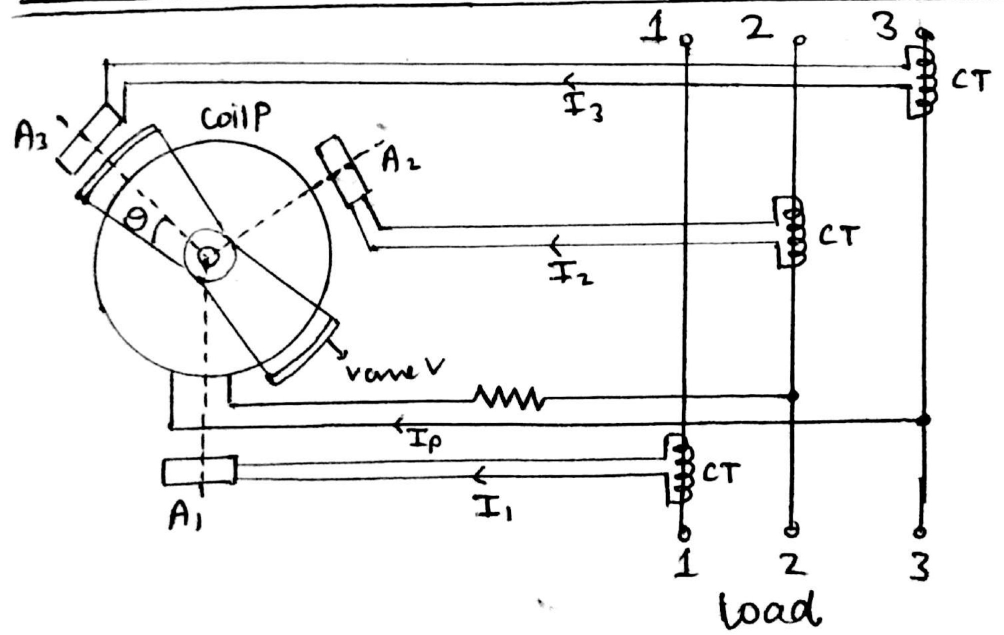
Hence at equilibrium condition, torque on moving coils will be same =>

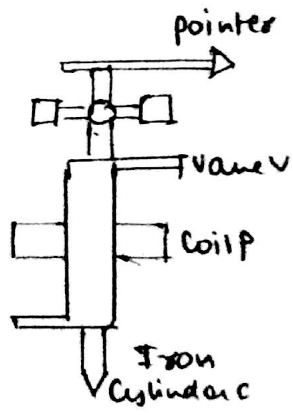
$$T_A = T_B$$

$$\Rightarrow \boxed{\theta = \phi} \quad \text{(on solving)}$$

$\therefore$ The angular deflection of the pointer will correspond to the phase angle of the circuit and hence power factor can be calculated.

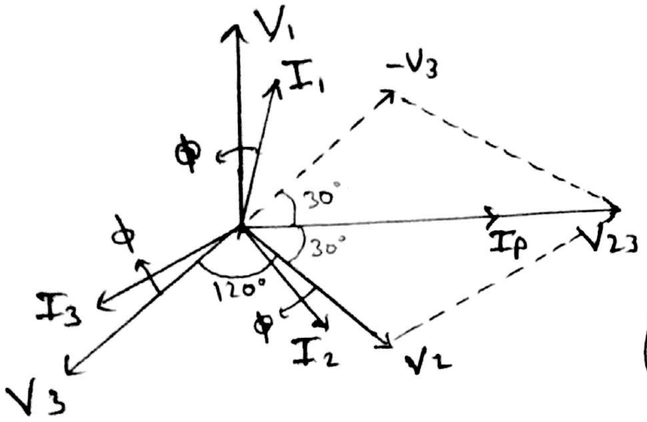
III Rotating Field Moving Iron Power Factor Meter.





A_1, A_2, A_3 are fixed coils placed 120° apart, connected to phase supply line 1, 2 and 3 respectively. Current T of are used to step down the current flowing in supply lines.

Coil P is placed, in series with a high resistance, across any two phase supply lines.
Due to alternating supply, a changing or rotating flux is generated. Flux produced by I_p in coil P, interacts with flux produced by coils A_1, A_2 and A_3. Hence a deflecting Torque is produced in the moving system.

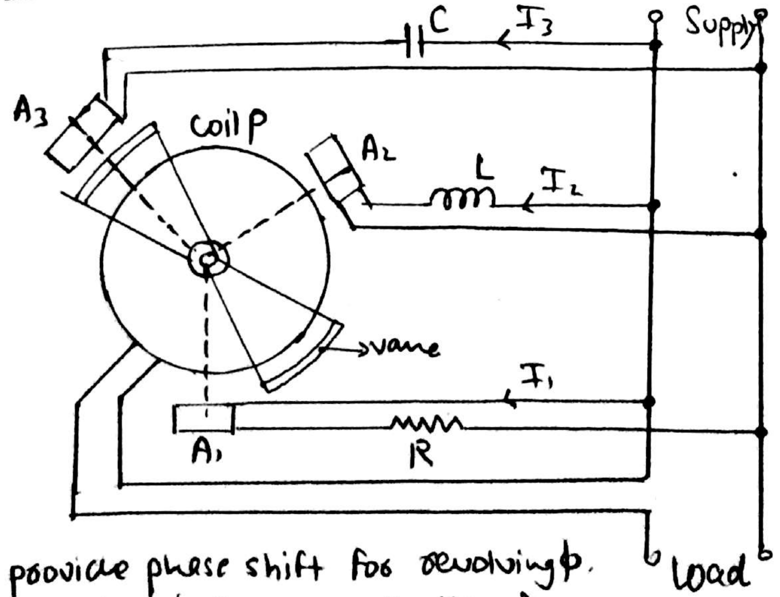


$$T \propto I_1 I_p \cos(90-\phi)$$
$$\propto I_2 I_p \cos(30+\phi)$$
$$\propto I_3 I_p \cos(150+\phi)$$

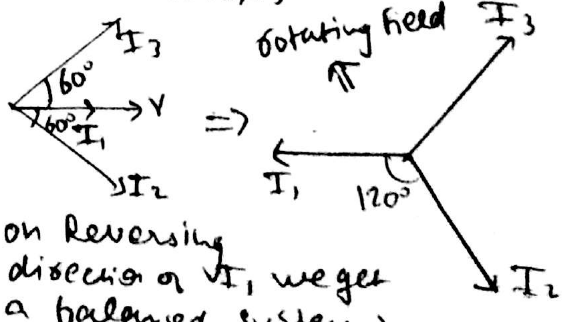
(deflection depends on phase angle b/w current in main line and PC)

Now, The Torque on Moving system is perfectly balanced by opposing forces, we get $\boxed{\theta = \phi}$.
→ No need of controlling Torque
→ pointer remains at occupied position.
Hence deflection of the moving part is a direct measure of phase angle b/w line current and the corresponding phase voltage.

IV Single Phase Moving Iron Power Factor Meter



R, L, C components are connected to provide phase shift to I_1, I_2, I_3

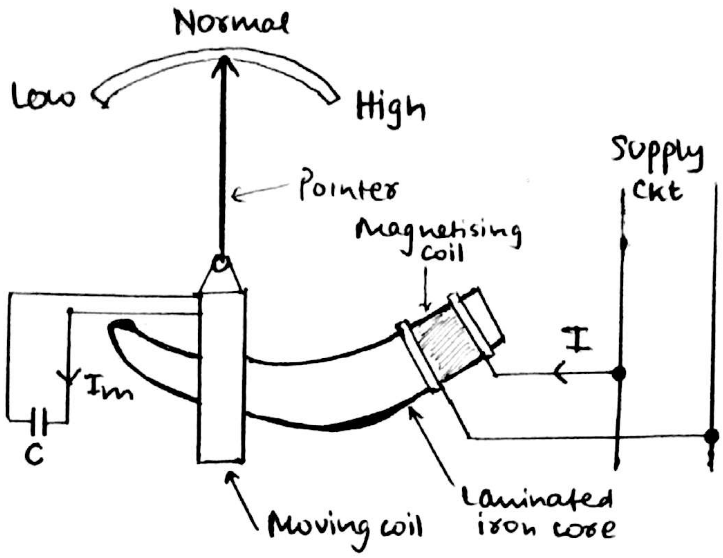


(on Reversing direction of I_1 we get a balanced system).
Hence revolving Field is produced.
(rest working is same as 3φ Moving Iron PF Meter.)

R, L, C provide phase shift for revolving φ.
(Rest operating is same as earlier).

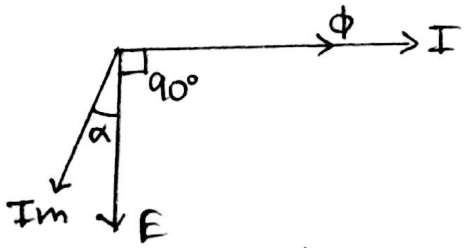
FREQUENCY METERS

Electrical Resonance Type Frequency Meter



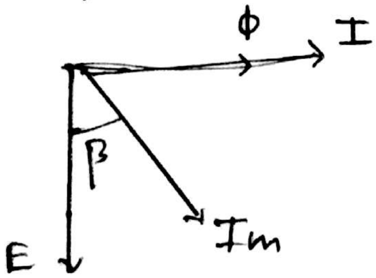
In this setup, a fixed coil is connected accross the supply whose frequency has to be measured. This is called the magnetising coil. This coil is 'tapered'. The Moving coil is pivoted over the iron core, and terminals of moving coil are connected accross a capacitor.

Operation →



In this case, circuit of moving coil is assumed to be inductive.

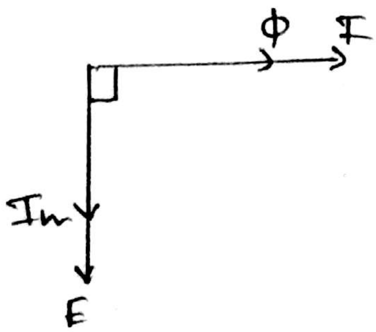
$T \propto I_m I \cos(90+\alpha)$



Now, the circuit of moving coil is highly capacitive.

$T \propto I_m I \cos(90-\beta)$

In this case Torque is 'opposite' due to phase



$X_L = X_C$ ∴ Resonance condition

No torque is produced

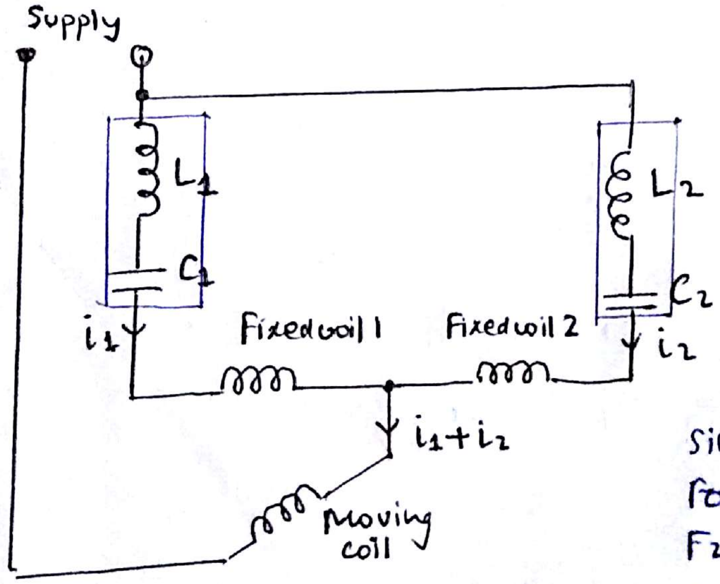
∴ $T = I_m I \cos 90° = \underline{0}$

At high frequency, X_L increases $(X_L > X_C)$
So a torque acts which tries to pull the moving coil towards equilibrium position by bringing $X_L = X_C$

At low frequency, X_C increases $(X_C > X_L)$
Torque is produce which changes position of moving coil according to equilibrium.
Hence, accordingly the movement takes place for the pointer.

→ Inductance of moving coil changes gradually with variation of position of iron core, hence instrument provides good sensitivity.

Electrodynamometer Type Frequency Meter

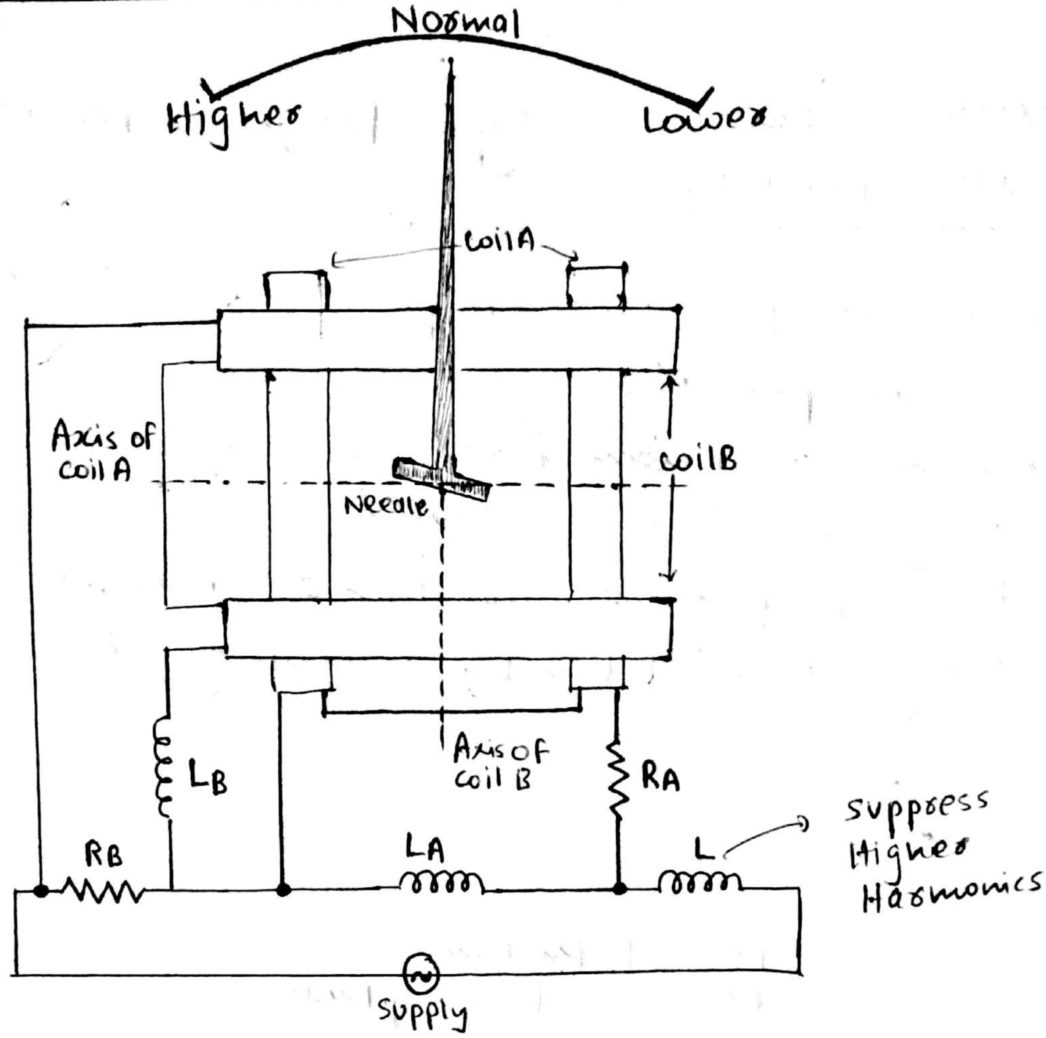


Fixed coil 1 is in series with L_1 and C_1 forming a resonant circuit of frequency f_1 which is slightly below lower end of instrument scale

Similarly, for Fixed coil 2, L_2 and C_2 form a resonant ckt of frequency F_2, slightly higher than upper end frequency of instrument scale.

The circuit of fixed coil 1 operates above resonant frequency with current i_1. The ckt of coil 2 operates below the resonant frequency, with current i_2. Hence one circuit is inductive and other capacitive due to leading and lagging nature of currents i_2 and i_1 respectively. Hence the resultant torque produced by i_1 and i_2 act in opposition, and are a function of frequency of applied voltage.
So, scale is calliberated in terms of frequency.

WESTON TYPE FREQUENCY METER



R_A is connected in series with coil A, and L_A is connected parallel accross coil A.
For coil B, R_B is connected parallel to the coil and L_B is in series with the coil.

Working / Operation

If frequency of supply is high, value of reactance offered by L_A i.e. X_{LA} increases, Voltage drop accross X_{LA} also increases, hence current accross this coil A increases.
In coil B, value of X_{LB} will increase but ∵ L_B is in series, the net current in coil B will derecrease.

⇒ $I_A > I_B$, field due to coil A becomes stronger and tendency of needle is to allign towards stronger field, hence it deflects itself with axis of coil A, i.e. toward left.
Likewise, if frequency decreases, opposite action takes place and the pointer deflects towards right.

Mechanical Resonance Type Frequency Meter (Vibrating Reed Type)

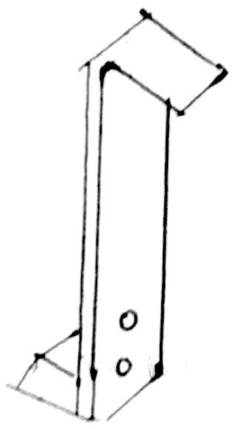
→ Reeds are thin steel strips

→ Reeds are placed alongside electromagnets

→ Electromagnet is connected across supply whose frequency has to be measured.

→ Natural frequency of vibration of reeds are different

→ Reading can be taken by resonating reed.



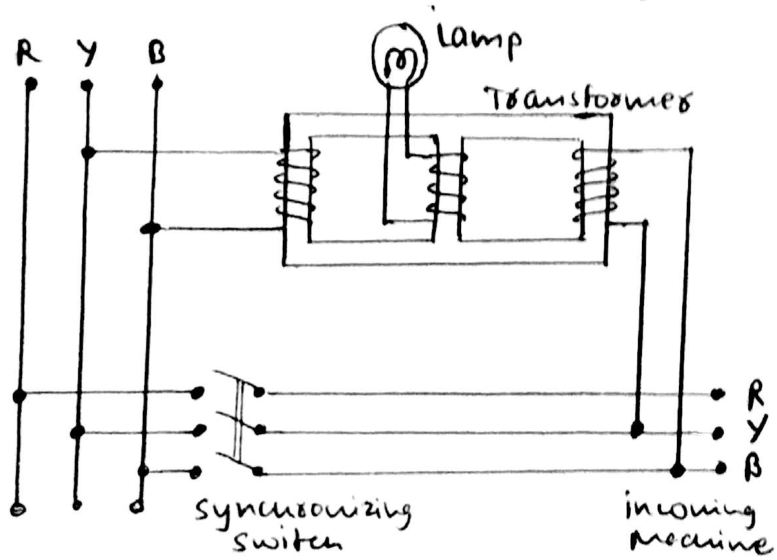
(Reed of Mechanical Resonance Frequency Meter)

SYNCHROSCOPES

A synchroscope is used to determine the correct instant for closing the switch which connects an alternator to power station busbars

The correct instant of synchronizing is when the busbar and incoming machine voltages

a) are equal in magnitude
b) are in phase
c) have the same frequency.



The windings of the outer limbs are exited by busbars and by limbs of incoming Machine. These outer windings produce flux. This flux induces voltage in the central limb. The resultant flux through the central limb will be equal to the sum of these two fluxes

When voltage of same phase, magnitude and frequency comes, max resultant flux will be present in the central limb, and hence the bulb will have maximum brightness and negligible flickering. This will be the correct moment of synchronising